

In MYP , command terms can be arranged from **simplest to more complex thinking** as follows:

Sample:

COMMAND TERM	MEANING	EXAMPLE SHOWING PROGRESSION Question	ANSWER
State	Give a short answer, fact, or name. No explanation needed	State one thing plants need to grow.	Plants need water to grow.
Select	Choose the correct item(s) from a list or options	Select the factor that helps plants make food. A. Soil B. Sunlight C. Rock D. Plastic	Sunlight
Outline	Give the main points or a brief description	Outline how sunlight helps plants grow.	Sunlight provides energy for photosynthesis. Plants use this energy to make food, which helps them grow.
Design	Create or plan something to solve a problem	Design an investigation to test how the amount	● Research Question: How does the amount of sunlight affect plant

	or answer a question	of sunlight affects plant growth.	growth? ● Independent Variable: Amount of sunlight ● Dependent Variable: Plant height ● Controlled Variables: Plant type, water, soil ● Method: Place identical plants in different amounts of sunlight and measure their height each week.
Progression:  State →  Select →  Outline →  Design The thinking becomes deeper at each step, moving from recalling information to creating an investigation.			

Design an **investigation** to test how the amount of sunlight affects plant growth

Materials

- 3 small same-sized plants
- 3 pots with the same soil
- Water
- Measuring ruler
- Labels
- Notebook/data table
- Sunny place, partial sunlight place, shaded place

Method

1. Take **3 same-sized plants** in similar pots.

2. Label them:
 - Plant A: Full sunlight
 - Plant B: Partial sunlight
 - Plant C: Shade/no sunlight
3. Keep the plants in their assigned places.
4. Give each plant the **same amount of water** every day.
5. Measure the height of each plant every 2 days.
6. Record the observations in a table.
7. Continue for 2 weeks.
8. Compare the growth of all three plants.

Safety Precautions

- Handle plants and pots carefully.
- Do not spill water on the floor.
- Wash hands after touching soil.
- Keep the pots in a safe place to avoid tripping.
- Use the ruler carefully; do not poke others.

ACTIVITY: IDENTIFY THE VARIABLES

QUESTION	INDEPENDENT VARIABLE	WHAT DO YOU CHANGE?	DEPENDENT VARIABLE	HOW DO YOU MEASURE IT?	CONTROL VARIABLES
1. How does the type of exercise affect heart rate?					
2. How does the amount of time spent studying affect test scores?					
3. How does temperature affect the solubility of salt in water?					
4. How does the design of a paper airplane affect the distance it flies?					
5. How does the type of breakfast affect students' energy levels throughout the morning?					
6. How does the amount of sunlight affect plant growth?					